

Sophix V3 DD Coherence Review And Hardening

Date: 2026-06-01

Status: Active review output

Purpose: record the cross-document coherence pass after all reviewed DDs and frontend/channel documents were generated.

1. Executive Verdict

The SOPHIX V3 DD set is directionally coherent and can guide implementation squads, provided the patched cross-map and index are treated as the current control documents.

The main problems found were not missing business areas anymore. They were:

- stale gap/register documents that still said modules were pending after they were created;
- duplicate or older module names that could make teams implement the same capability twice;
- older "future" references around cross-technology migration;
- overdesign risk from interpreting every worker/topic as a separate pod;
- overuse risk for Camunda and Drools in places where normal service code or scheduled jobs are enough.

2. Corrections Applied

Area	Correction
Global navigation	Rebuilt DD_INDEX.md as a true index and implementation guardrail document.
System traceability	Updated DD_CROSS-00 with late modules, frontend/channel links, technology migration, field audits, CVM, reporting, and USSD.
Runtime simplification	Added shared guidance for worker consolidation, Camunda use, Drools use, and reporting/read-model performance.
Performance/complexity assessment	Added Sophix_V3_Performance_Complexity_Assessment_20260601.md as the standalone assessment for module count, process complexity, simplification, phasing, and bottleneck risk.
MVP runtime baseline	Added Sophix_V3_MVP_Runtime_Simplification_Baseline_20260601.md as the binding first-release implementation baseline.
Process coverage status	Marked the original process coverage register as historical and added the current closed-gap mapping.
Gap reconciliation	Marked the original gap reconciliation as historical and added the current closed-gap mapping.
EM-02 duplicate	Marked Contractor_Team_Agent_Registry as superseded by Contractor_and_Staff_Registry.
Technology migration	Clarified that SUB-WF-MIGRATION-01 owns the business operation and FUL-10 owns technical cross-technology fulfillment.
WO contract	Updated the subscription-to-WO contract to reference current TECHNOLOGY_MIGRATION handling instead of a vague future migration flow.
Generated dev guide endpoint label	Renamed misleading Primary endpoint metadata to Endpoint scan reference across generated guides. The DD API section and execution flow now remain authoritative for command order.

3. Coherence Decisions

Topic	Decision
ASR/TCK	ASR DDs are ticket/service-request workflow satellites. TCK remains the ticket lifecycle engine.
Field audit	FA-01, FA-02, and FA-03 can be one deployed capability with audit_type, shared tables where practical, and separate rules/workflows only where needed.

Topic	Decision
Reporting	FE-APP-05 must use REP-01. Browser dashboards must not query operational DBs directly.
USSD	USSD is a channel adapter and should hold only session/menu/audit state.
CVM	Start with rule-based segmentation and queues. Do not introduce an ML platform in v1 unless a concrete business requirement appears.
Workers	Worker topic names are logical contracts. Consolidation is expected in a medium-budget deployment.
Camunda	Use for long-running orchestration, timers, compensation, and approvals. Do not use for simple CRUD or deterministic scans.
Drools	Use for configurable business policy. Keep deterministic validation in service code.
Cache	Cache improves reads but does not become transactional truth for state-changing decisions unless a DD explicitly says so.

4. Overdesign And Performance Risks

Risk	Why it matters	Hardening recommendation
One pod per worker topic	Too many deployments become hard to monitor and expensive to operate.	Start with grouped worker apps and per-topic metrics. Split only when load/security/team ownership requires it.
Camunda for everything	BPMN becomes noisy and slows simple feature delivery.	Put only real workflows in Camunda. Use scheduled commands/jobs for deterministic background work.
Drools for fixed validation	Rules become harder to test and debug when they duplicate code constraints.	Use Drools only for configurable policies and decision tables.
Frontend fan-out for dashboards	Operational APIs and databases can become overloaded.	Use REP-01 mart/API for dashboard totals and exports.
Duplicate read models without ownership	Teams may trust different sources for the same state.	One source-of-truth module per object; read models must declare freshness and reconciliation behavior.
Cross-tech migration duplicate state	Subscription, fulfillment, WO, and OSR could each invent a migration status.	Use one subscription operation id/correlation id; each module owns its local state only.
Too many late DDs as separate services	Small capabilities become operational overhead.	Deploy related low-volume capabilities together where ownership and scaling allow it.

5. Remaining Work Before Implementation Freeze

Item	Required before
Regenerate PDFs and the review zip from patched markdown	Formal review pack distribution.
Refresh HLD and process-map PDFs to match the final DD set	Executive/architecture review.
Produce OpenAPI contracts for the first implementation wave	Parallel frontend/backend development.
Turn DD_QA-01 scenarios into automated tests	SIT/UAT readiness.
Confirm optional naming cleanup for FUL-10/field-audit/ASR codes	Final baseline publication.

6. Implementation Readiness Statement

After this hardening pass, the design can be used for squad planning if teams follow these controls:

- start from DD_INDEX.md and DD_CROSS-00;
- implement the canonical owner only when duplicate historical DDs exist;
- build module APIs before frontend flows that depend on them;
- keep OpenAPI, contract tests, and E2E scenarios synchronized with each implemented feature;
- treat older gap reports as historical unless their top update section says an item remains open.